15.3 Constructor overloading

Function overloading can be applied for constructors, as constructors are special functions of classes .A class can have more than one constructor with different signature. Constructor overloading provides flexibility of creating multiple type of objects for a class.

Illustration 15.3 Constructor overloading

```
#include<iostream>
using namespace std;
class add
{ int num1, num2, sum;
public:
add()
{         cout<<"\n Constructor without parameters.. ";
num1= 0;  num2= 0;  sum = 0; }
add ( int s1, int s2 )
{ cout<<"\n Parameterized constructor... ";
 num1= s1;  num2=s2; sum=0; }
add (add &a)
{ cout<<"\n Copy Constructor ... ";
 num1= a.num1;
 num2=a.num2;
 sum = 0; }
void getdata()
{ cout<<"\nEnter data ... "; cin>>num1>>num2; }
void addition()
{ sum=num1+num2; }
void putdata() {
cout<<"\n The numbers are..";
cout<<num1<<'\t'<<num2;
cout<<"\n The sum of the numbers are.. "<< sum; } };
int main() {
add a, b (10, 20) , c(b);
a.getdata();
a.addition();
b.addition();
c.addition();
cout<<"\n Object a : ";
a.putdata();
cout<<"\n Object b : ";
b.putdata();
cout<<"\n Object c.. ";
c.putdata();
return 0; }
```

Compiler identifies a given member function is a constructor by its name and the return type.

Chapter 15 Page 249-259.indd 252 3/24/2020 9:23:24 AM